

Arnav Patil

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EDUCATION

University of Toronto

Sept 2023 – Apr 2028

BASc. in Computer Engineering with PEY Co-op

Toronto, ON

- **GPA: 3.58/4.00** with recognition on Dean's Honours List
- **Minor in Engineering Business**

EXPERIENCE

Digital IC Design Intern

May 2026 – Present

Marvell Technology

Toronto, ON

- Will be responsible for block- and chip-level RTL design and verification for Marvell's 10 – 224 Gbps SerDes IP.

Teaching Assistant

Sept 2025 – Dec 2025

University of Toronto

Toronto, ON

- Facilitated tutorials for **APS100 Orientation to Engineering**, reinforcing concepts through discussion.
- Provided mentorship on academic strategies, course planning, and career development to first-year students.

FPGA Engineer Intern

May 2025 – Aug 2025

Intelligent Sensory Microsystems Lab

Toronto, ON

- Developed FPGA-based modules for novel CMOS time-of-flight sensors in Prof. Roman Genov's **ISML** group.
- Achieved 800 Mbps sensor readout throughput by designing high-speed RTL modules on Artix-7 FPGAs.
- Interfaced with peripherals by implementing SPI and I2C controllers in Verilog, enabling full sensor configuration.

PROJECTS

Multi-Objective Smooth Analytical Placer (MOSAIC) | [🔗](#)

- Submission to the **2026 Macro Placement Challenge** co-hosted by Partel and Hudson River Trading.
- Achieved a 10.3% improvement on RePlace on IBM ICCAD04 benchmarks, averaging a 99 s/bench runtime.
- Implemented differentiable approximations for the TILOS MacroPlacement evaluator to minimize cost in Python.

Deep Learning for Analog Circuit Design (SpecNet) | [🔗](#)

- Designed a neural network pipeline to automate transistor-level analog circuit sizing from target specifications.
- Achieved 98% accuracy within 5% of optimal dimensions, greatly reducing number of SPICE simulations required.
- Automated dataset generation by parsing LTspice simulation logs into structured ML training data using Python.

DE1-SoC Arcade Game | [🔗](#)

- Implemented a blackjack game in Verilog on an FPGA, managing FSMs for game logic and player interaction.
- Simplified hardware peripheral control from top-level modules by writing API-like wrapper RTL over Altera IP.
- Fixed version compatibility issues in provided IP by writing a Python script to re-format memory files (.mif).

Nios-V Ultrasound System | [🔗](#)

- Integrated an ultrasonic sensor and servo motor into the FPGA-based Nios-V soft processor via GPIO ports.
- Designed & implemented BJT pull-up/pull-down networks for safe voltage conversion between FPGA and sensor.
- Implemented timer-based polling in C to calculate object distance from 20 cm to 2 m within 2 cm precision.

Spatial Audio Engine on ARM Cortex-M4 | [🔗](#)

- Built a real-time DSP pipeline on a Cortex-M4 MCU, processing stereo audio streams end-to-end in bare-metal C.
- Implemented fixed-point arithmetic to eliminate floating-point overhead, maximizing computational throughput.
- Engineered dynamic stereo audio by computing channel gain and phase delay to perceive an accurate position.

COURSEWORK

Languages & Tools: C/C++, (System)Verilog, Python, RISC-V Assembly, Quartus Prime, Vivado, Virtuosio, gem5

Hardware Courses: Computer Architecture, Digital & Analog Electronics, Computer Hardware, Signal Processing

Software Courses: Operating Systems, Embedded Programming, Machine Learning, Microarchitecture Simulators